

Launch Readiness — The First 1,000 Clients

50pick · 50pick.tz · brief of 3 September 2026, **re-verified line by line and updated 4 September 2026** · production commit 011733eb . Every figure re-measured on the live system; two claims in the original brief were wrong and are corrected below.

What changed on 4 September (backups-first, by Ali's decision, to keep early costs minimal).

- ✓ **Backup schedules are ON** — daily + weekly + monthly snapshots of the database volume, daily + weekly of the cache volume, plus a named baseline snapshot. Cost: cents.
- ✓ **The backup alarm now arrives by itself** — if the nightly backup fails, stalls, or silently stops running, every administrator gets an email and an in-app alert within a day. The "eleven nights unnoticed" failure can no longer repeat quietly.
- ✓ **The status page can now say "not ready"** — it answers 503 when the database is unreachable, proven both ways by an automated test, live on production.
- ✓ **A spending alarm exists** — Railway emails at \$75/month. No hard cap (a hard cap would switch production off).
- ✓ **A live credential found committed in the repository was removed** (rotate it in Postmark).
- ⊖ **One command is still yours to run** — enabling point-in-time recovery: `railway postgres pitr enable --service Postgres` (~\$1–3/mo, a brief database restart, and the recovery window only starts counting once it is on).

1 · What we pay now

Service	Per month	What it is for
Railway — 50pick hosting	\$23.57	The application, database and cache
Railway — five other projects	\$8.24	Amanat Watan, Intercontinental Glass, LMC, two others
Anthropic — AI	\$50.75	Writing markets and auto-settling results
Cloudflare R2	~\$1.00	Customer ID documents and backups
Postmark email / Sentry / SMS	\$0.00	Free tier or not yet sending
Twelve Data price feed	not measured	Billed outside Railway — needs confirming
Total today	≈ \$84	+ cents for the new volume snapshots; +\$1–3 when PITR is enabled

The two facts that decide where the money goes — both re-confirmed.

The AI bill is bigger than the hosting bill (\$50.75 vs \$31.81) and 99.98% of it does not depend on customer numbers.

Memory is 87% of the 50pick hosting bill — the cheapest saving available, parked for phase 2 at your instruction.

2 · What we will pay with 1,000 clients

Unchanged from the original brief (re-checked): **budget \$200 per month**, range \$155–290, dominated by hosting growth, the AI clock, and ~\$35–45 of SMS once the Selcom contract lands.

3 · The readiness list — verified state, 4 September

	Item	Cost	Verified state
●	Alerts that arrive by themselves	\$0	DONE, live. A daily watchdog inside the platform emails and pings every administrator when the nightly backup is missing, failed, unverified or stale — including the case where the schedule silently stops running, which no CI-side alert can see. Proven by 24 automated checks including the "must stay silent when healthy" direction.
●	Volume snapshots	~cents	DONE, live. Railway-native daily/weekly/monthly snapshots of the database volume (kept 6/27/89 days), daily/weekly of the cache, plus a named baseline. A second recovery mechanism sharing nothing with the nightly backup.
●	The status page can fail	\$0	DONE, live. <code>/api/health</code> answers 503 when the database is unreachable or unmigrated; HEAD agrees with GET; proven RED first by an automated suite, then green against production. (The original brief said this was "proved both ways" on a branch — it had no automated proof, and its green case exercised the one branch designed never to fail.)
●	Continuous database backup (PITR)	\$1–3/mo	One command, yours: <code>railway postgres pitr enable --service Postgres</code> . Brief database restart on enable; the ~4-week restore window starts only after it is on — every day it stays off is a day it does not cover. Restoring creates a NEW sibling database and never touches the live one. Cuts possible data loss from ~24 hours to minutes.
●	Turn on SMS	\$35–45/mo	Commercial step stands — but the original justification was wrong. The brief said "no member of the public can register today". Re-verified in code: registration is phone + password and works with SMS off; the code-by-text path is written but deliberately bypassed until the contract lands. SMS is needed for phone verification and codes — a compliance and quality item, not a sign-up blocker. The Selcom adapter has never sent a real message: first send needs a live-handset test.
●	Two-factor on the admin console	\$0	Open (campaign item E-255), and one fear in the file is disproven: the code contains a forced-enrolment flow (an admin without a code app is sent to the setup page, which is exempt from the gate — no lockout is possible by design). But that flow has never been driven live, so the order is: prove it locally, flip the flag, walk one QA admin account through enrolment on production, then every admin enrolls at next login.
●	Updates without downtime	\$0	Authored, one command from live. The deploy gate (health check + 60s overlap + 30s draining) is written as reviewable infrastructure-as-code on branch <code>launch-1k-phase2</code> ; Railway's

			preview confirms it changes exactly four fields and destroys nothing. Discovery that mattered: the config file format the original plan assumed (<code>railway.json</code>) has been deprecated by Railway and is silently ignored — a deploy carrying it went live unconfigured while looking green.
●	Payouts on large markets	\$0	Re-measured today on a scratch database, and the original number was right: 26 database round-trips per winner, one at a time, inside a 30-second limit — settlement dies at ~576 winners at Railway's own network distance (the brief said ~660). Worse than the brief knew: a failed settlement now rolls back completely and retries forever without progress, so an over-limit market freezes permanently. The batch fix is designed (all winners in a handful of set-based statements, same all-or-nothing safety) and is phase 2's first code item, with a 1,000–5,000-winner proof harness already in the repo.
●	Cache the busy screens	\$0	Mostly already true — the brief's premise was stale. Re-derived: the results archive has a 5-minute cache, card charts are one batched query per board, platform stats are cached 60s, and board reads are indexed scans over ~93 rows. The one real gap — the leaderboard recomputing ~100 queries per viewer — is fixed on branch <code>launch-1k-phase2</code> (computed once per 20s for everyone). No new infrastructure, \$0.
●	Move servers to Amsterdam	\$0	Parked for phase 2 (your call: minimal changes first). Confirmed live: no CDN in front, ~330–490 ms first byte from distance alone, halving available. Needs a maintenance window — the app, database and cache move together, maintenance mode on (withdrawals stay open).
●	Enlarge the database disk	\$0 now	919 MB of 5 GB, ~14 MB/day, ~6 months of headroom. Discovery: enlarging is dashboard-only — one click for you, no API exists. Costs nothing until used.
●	Recovery procedure and rehearsal	\$0	The backup half is now written into the backup runbook (watchdog, snapshot layers, PITR enable/verify/restore steps). Still owed: one timed restore rehearsal at production size — the platform's recovery time remains unmeasured until then. Scheduled right after PITR is on.

Three things only you can do: run the PITR command · rotate the exposed Postmark token · sign the Selcom SMS contract. Everything else on this list is either done, one `railway config` apply away, or parked in phase 2 at your instruction.

4 · Decisions taken

Question	Decision
Sizing	1,000 registered · 100–200 online at peak (unchanged)

Scope, September	Backups and alarms first, minimal cost (Ali, 4 Sep). Deploy gate, payout batching, Amsterdam, memory cap, 2FA flip = phase 2, staged and documented on <code>launch-1k-phase2</code>
Run several copies?	No — re-confirmed with stronger evidence: a second copy would silently defeat self-exclusion enforcement (a regulatory control), not just ops conveniences
Selcom SMS contract	In progress; switch is env-only + one code re-route, first send needs a real handset

Part Two · Technical detail

MEASURED 3 SEPTEMBER · RE-DERIVED 4 SEPTEMBER 2026

Architecture as deployed (unchanged 4 Sep except where marked)

```
Players (Tanzania)
|   no CDN, no WAF (verified: no cf-ray; edge cdg1 -> origin us-west2)
v
Railway edge ----> ORIGIN: us-west2 (California) -- all three services
|
+-- 50pick          Next.js 16, RAILPACK          1 container
|                   health endpoint CAN FAIL (503) ✓ · Railway gate pending `config apply`
+-- PostgreSQL 18.6  5 GB disk (919 MB) · volume snapshots D/W/M ✓ · PITR pending 1 cmd
+-- Redis 8.2.1      connected, cross-container · volume snapshots D/W ✓

External services: Selcom LIVE · Postmark LIVE · R2 LIVE · Sentry LIVE ·
Anthropic LIVE · Twelve Data LIVE · SMS NOT DELIVERING (provider=console)
```

26

DB round-trips per winner
(measured 4 Sep)

~576

winners where settlement
dies (Railway RTT)

24/24

backup-watchdog proof suite

329/0

1 Hz probe ok/fail through a
live deploy

Corrections to the 3 September brief — each re-measured, not argued

- **"No member of the public can register today" — false.** Registration is phone + password (auth/register/actions.ts); the OTP path is preserved-but-bypassed. SMS remains a launch item for phone verification, not for sign-up.
- **"Proved both ways" on the health fix — overstated.** No automated test existed; the manual green exercised the no-database bypass branch. Now: test:health-readiness proves RED (8 assertions, dead DB → 503, no host leak) before trusting any green, and the true-positive branch is proven against production.
- **The "~660 winners" figure was right** — re-measured at ~576 at same-region RTT (26 round-trips/winner, current code, scratch DB at production schema). What the brief missed: the failure mode is now a permanent freeze, not a slow grind (full rollback + 5-min retry with zero progress). Detection exists ("Ready to settle" > 0 on /admin/system).
- **"Nothing is cached" — stale.** Five server-side cache layers already exist; the genuine gap (leaderboard) is fixed on the phase-2 branch.
- **The branch launch-1k-readiness must never be merged** — it is 33 commits behind and carries another session's swept-in work (its own tracker records the git add -A incident). Its one valuable code change was cherry-picked onto main with tests; its docs were rewritten fresh (this file, RAILWAY-LIVE.md).
- **Two API paths in the ops doc rotted within a day:** the CLI user token no longer answers billing queries (workspace token now), and railway.json is deprecated and silently ignored (.railway/railway.ts + plan/apply is the home).

Money safety — built, tested, and not rebuilt (re-verified in code 4 Sep)

- Maintenance mode pauses bets + deposits, withdrawals stay open · payment kill switches per network and direction · deposits settled from a signed re-query, exactly once · dead Redis cannot break a bet · no idempotency key → no automatic retry · 200 simultaneous bets: zero drift · the Up & Down safety net is

independent of the game's off switch — **every claim confirmed at file:line in the settlement path review.**

- **New since the brief:** backups are now watched by an alert that arrives (bell + email to every officer), and the volume layer snapshots independently of the nightly.

Recovery position

Question	3 Sep	4 Sep	After PITR + rehearsal
Data loss window	up to 24–27 h	24 h nightly + daily volume snapshots	minutes
Told of backup failure?	No	Yes — watchdog email + bell	unchanged
Written procedure	none	backup half in BACKUP-RUNBOOK.md	one-page DR runbook
Time to restore	never measured	never measured	measured and recorded

Order of work — updated

#	Step	State / proof
1	Volume snapshot schedules + baseline	✓ live — read back via API (D/W/M, retentions 6/27/89 d)
2	Backup watchdog	✓ live — 24/24 incl. silent-when-healthy; re-announces after each deploy
3	Health endpoint can fail	✓ live — RED 8/8 proven before green; 503 + not-ready on prod verified impossible-to-fake
4	Spend alarm	✓ live — \$75 soft, email, no hard cap
5	Enable PITR	🛑 Ali, one command ; then verify archiver + first base backup
6	Deploy gate (healthcheck/overlap/draining)	staged on <code>launch-1k-phase2</code> ; merge + <code>railway config apply</code> ; prove with 1 Hz probe <i>and</i> a manifest read (the probe alone lied once already)
7	Batch large-market payouts	phase 2 first code item — baseline measured, harness sweeps to 5,000 winners
8	Timed restore rehearsal → DR runbook	after PITR; record the wall-clock number
9	Amsterdam · memory cap · 2FA flip · SMS	phase 2 / commercial, in that order of ceremony

Original brief 3 September 2026 (commit `78c3876e`); re-verified and updated 4 September 2026 against production commit `011733eb`. Sources: Railway GraphQL API (workspace token), Railway CLI v5.49, the production database via read-only ops probes, timed requests to `50pick.tz`, a scratch PostgreSQL 18 at production schema for the settlement measurement, and the repo's own proof suites. Still unmeasured: the Twelve Data subscription. Rebuild: `node docs/runbooks/mkpdf.mjs ../launch-1k.html ../50pick-launch-1k.pdf`.